

Sri Lanka Institute of Information Technology



Assignment 01 CSSE_2025_Y3_NU_WE2

Case Studies in Software Engineering – SE3070

B.Sc. (Hons) in Information Technology- SE

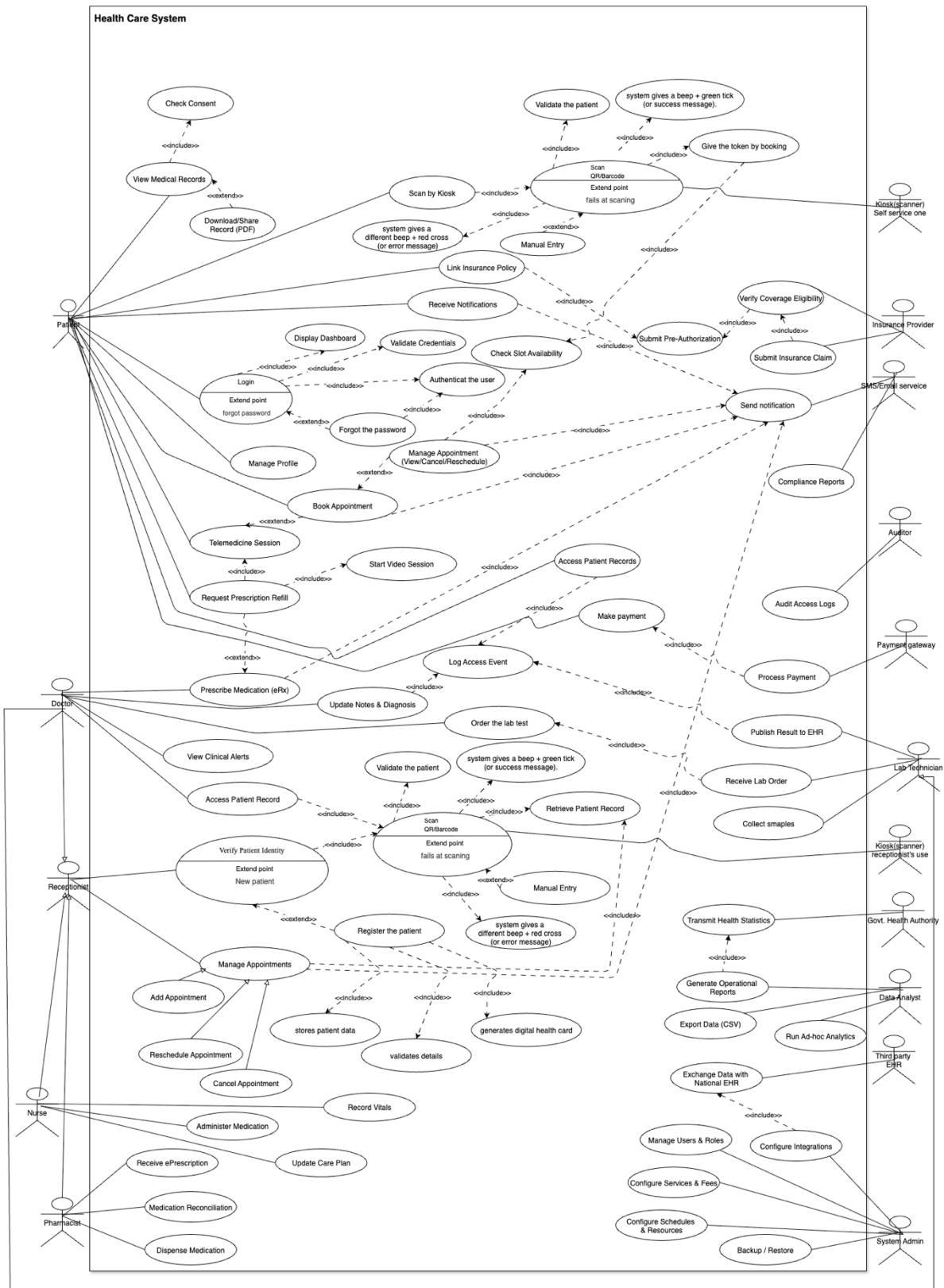
Group ID: CSSE_2025_Y3_NU_WE2

Campus: NorthenUni

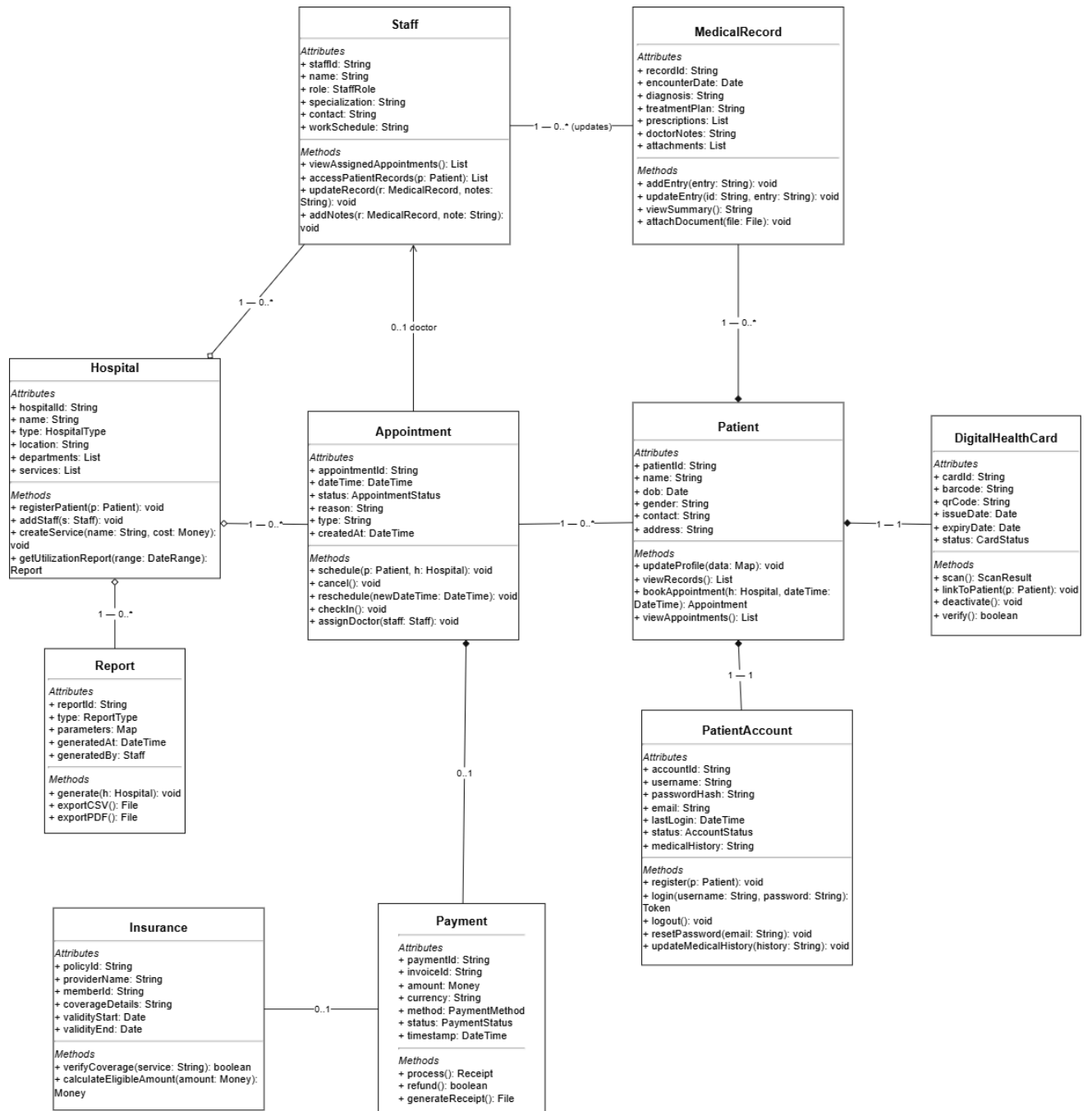
Group Members:

	Registration No.	Name
1	IT23276628 (Leader)	V. SHIVAANICA
2	IT23315778	K. HARUSHANAN
3	IT23335332	S. KAJANAN
4	IT23297586	S. NIRANJAN

High-Level Use Case Diagram



Class Diagram



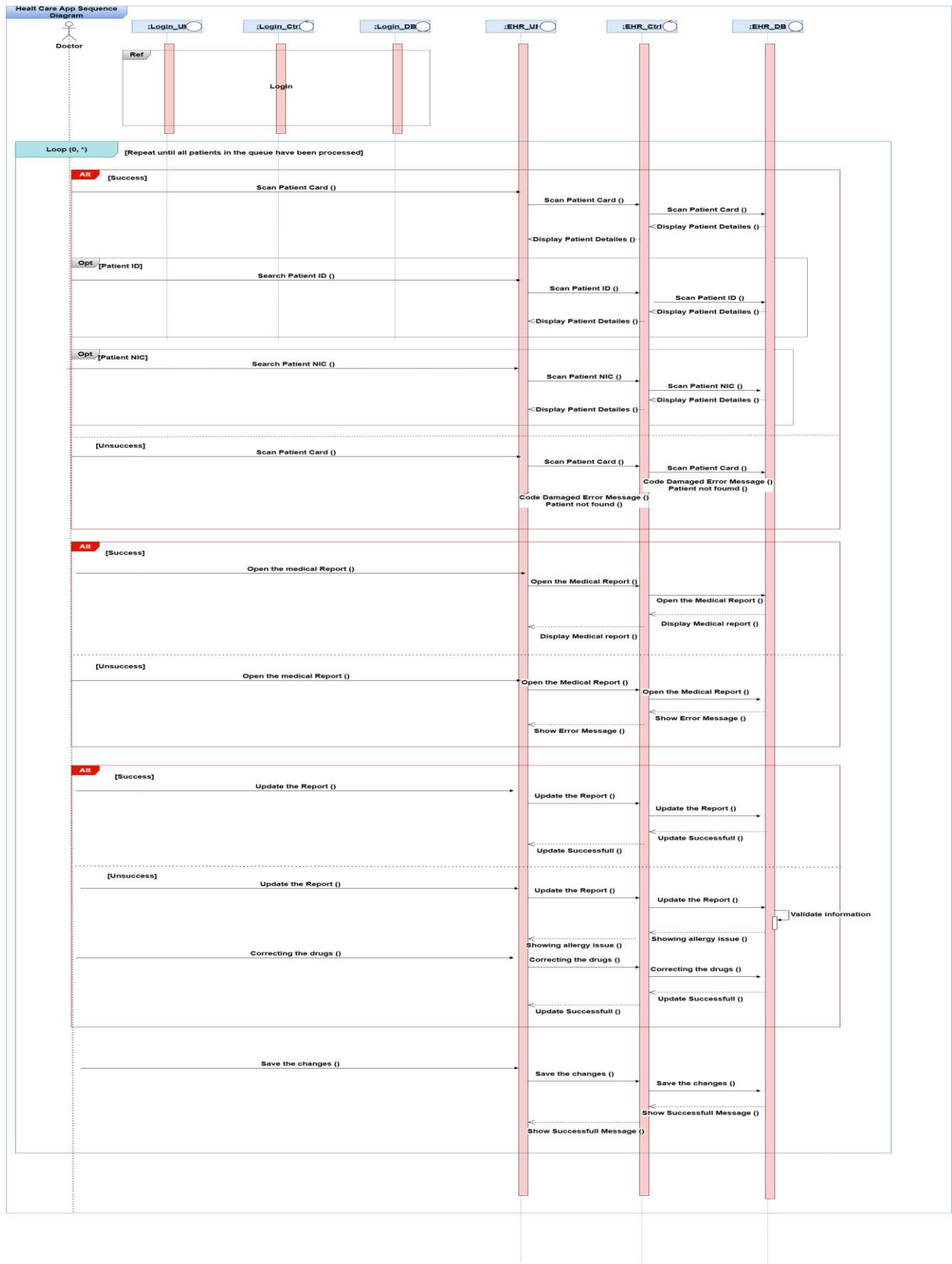
Detailed Use Case Scenario - Doctor Accesses Patient Card & Updates Record

Use case ID	EP-DOC-01	
Use case Name	Doctor Accesses Patient Card & Updates Record	
Summary	Authorized doctor scans/opens an electronic health card of a patient, looks at the most recent medical record, and updates clinical data (visit notes, vitals, diagnoses, medications), saving changes with an audit trail.	
Priority	5 (1=lowest priority, 5=highest priority)	
Preconditions	1) Patient has valid digital health card (barcode/QR or app). 2) Doctor is authenticated and has the right role-based permissions. 3) Patient records exist within the system.	
Post Condition	Update patient records are saved securely with timestamp, Doctor ID, and audit records; appropriate stakeholders (pharmacy, lab) can see authorized updates in real time.	
Primary actor	Doctor	
Trigger	Doctor scans the patient's electronic card (workstation or kiosk) or searches and selects the patient profile from the system.	
Main Success Scenario	Step	Action
	1	Doctor logs into the hospital EHR system with legitimate login credentials.
	2	The doctor selects "Find Patient" and scans the patient's electronic health card (barcode/QR) or searches by NIC/Patient ID.
	3	The system locates the patient profile and displays a confirmation (name, DOB, photo/identifier) to prevent misidentification.
	4	Doctor opens the patient's medical record from the results.
	5	The system displays the patient's complete record (demographics, history, allergies, current medications, recent visits, labs, imaging).
	6	Doctor navigates to the current encounter/visit (or creates a new encounter when necessary).
	7	Doctor enters the updates: clinical notes, vitals, diagnosis codes, orders (labs/imaging), prescriptions, and follow-up plan.
	8	Doctor reviews the updates and clicks "Save/Sign."
	9	System verifies required fields, searches for contraindications (Allergy drug interactions) and requests confirmation if required.

	10	System saves changes, writes audit log (who, what, when), and refreshes downstream views (Pharmacy queue for prescriptions).
	11	System shows success message and returns to patient record overview with changed entries displayed.
Extensions	Step	Branching Action
	2a	Card not readable or scanner error, System requests retrieve scan or manual search; marks failed scan attempt.
	2b	No patient found to match, System shows "Record not found" and offers re-search or escalate to registration.
	3a	Potential misidentification (mismatched populations), System notifies doctors and denies access pending verification.
	4a	Denied access (insufficient privileges), System shows authorization error and suggests calling an administrator.
	6a	Active encounter not found, System requests open/create new encounter context before updates.
	7a	Required data missing (Diagnosis for prescription), System shows required fields and denies save until completed.
	7b	Drug-allergy or drug-drug interaction detected, System displays safety alert with choices; Doctor must accept/adapt.
	7c	Offline/Network fault during editing, System offers temporary local draft store and auto-sync on reconnection.
	8a	Doctor aborts without saving, System discards unsaved changes (or saves a draft on user option).
	9a	Validation fails (format/values), System highlights fields with inline errors; save is prevented until corrected.
	10a	Save conflict (record changed by some other user), System shows a merge/conflict dialog with "Review changes," "Overwrite," or "Reload."
	10b	Downstream service is not available (pharmacy interface is down), System saves changes to EHR, queues outbound messages for retry, and notifies the doctor.
Open issues	1	Should consent prompts be required for sensitive topics (Sexual health, mental health) before the doctor can view/update?
	2	Do we have records existence context for ALL edits, or do we allow quick edits with later reconciliation?

	3	What is the length of the session timeout before auto-save, or auto-lock is triggered?
	4	Does the system automatically alert assigned nurse/pharmacist to update to specific fields (STAT orders)?
	5	How is versioning and rolling back done if an invalid update must be rolled back?

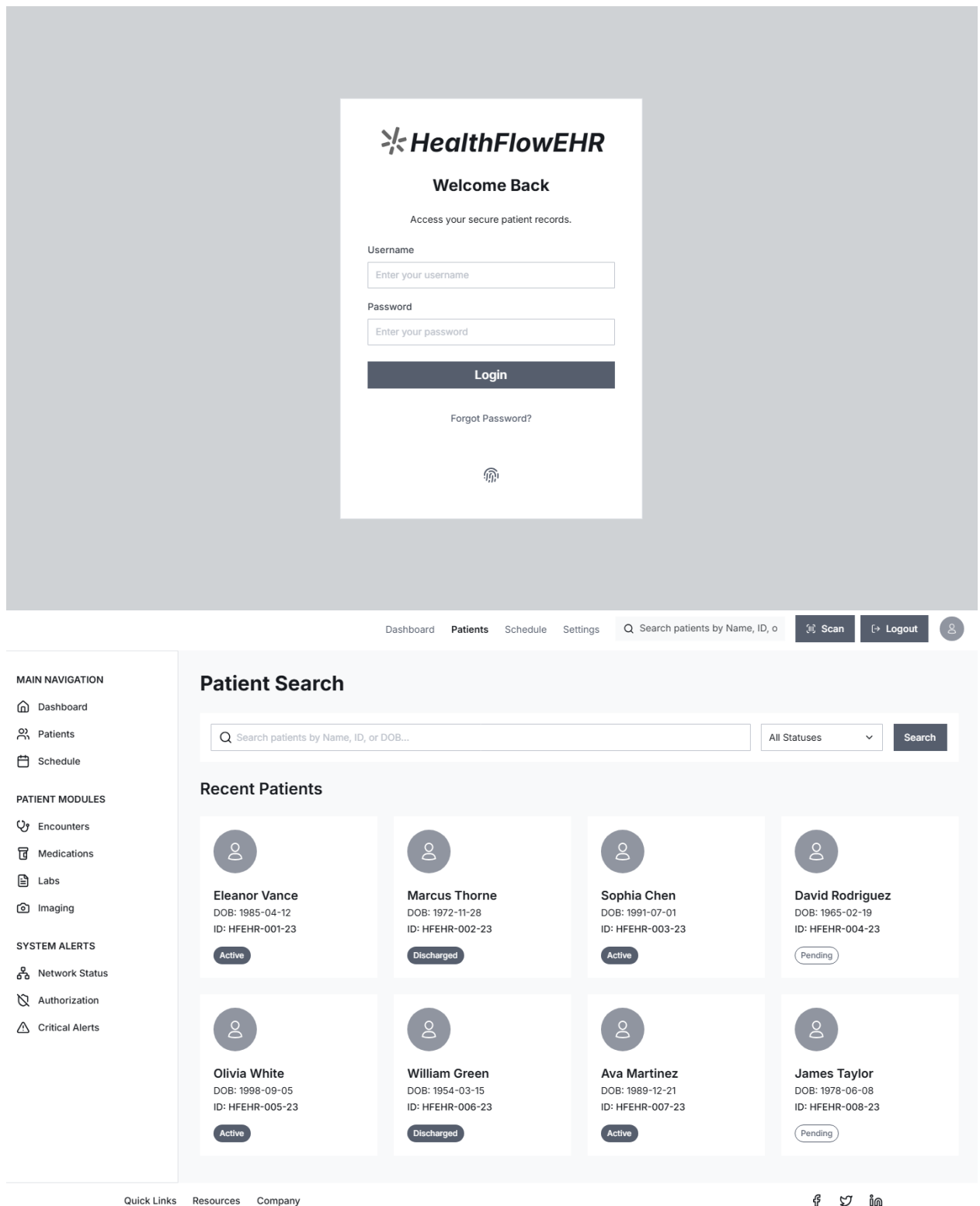
Detailed Sequence Diagram



Storyboard



Low-fidelity Wireframes





MAIN NAVIGATION

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PATIENT MODULES

[Encounters](#)[Medications](#)[Labs](#)[Imaging](#)

SYSTEM ALERTS

[Network Status](#)[Authorization](#)[Critical Alerts](#)

John Doe

DOB: 1975-08-20

Gender: Male

Patient ID: HD75-08-20-001

Attending Physician: Dr. Emily Smith

Demographics

Allergies

Medications

Labs

Imaging

Visits

Demographic Information

Full Name: John Doe

Date of Birth: 1975-08-20

Gender: Male

Patient ID: HD75-08-20-001

Contact: (123) 456-7890

Address: 123 Healthway, Medical City, 00000

Emergency Contact: Jane Doe (Spouse) - (987) 654-3210

DashboardPatientsScheduleSettings

Search patients by Name, ID, o

ScanLogout

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Critical Alerts

Anya SharmaID: HFEHR00123DOB: 1985-04-15

Back to Patient Record

Clinical Notes (SOAP)

Subjective

Patient reports worsening cough and shortness of breath for 3 days. Denies fever or chills. Has history of asthma, usually well-controlled. Last inhaler use yesterday morning with minimal relief.

Objective

Vitals: BP 120/80, HR 88, RR 20, Temp 98.6°F, SpO2 96% on room air. Pause: Bilateral wheezing, prolonged expiration. Cardiac: Regular rate and rhythm. ENT: Clear. No edema.

Assessment

Acute exacerbation of asthma. Possible respiratory infection contributing. Stable condition, no immediate distress.

Plan

1. Nebulizer treatment with Albuterol in clinic.

2. Prescribe oral prednisone for 5 days.

3. Continue current asthma controller medication.

4. Return to clinic in 3 days or sooner if symptoms worsen.

5. Patient education on inhaler technique and symptom management.

Vitals

Blood Pressure

120/80

Heart Rate

88

Temperature

98.6

Resp Rate

20

SpO2

96

Diagnosis Codes

Search or enter ICD-10 code (e.g., I10, E11.9)

+ Add

J45.901 - Unspecified asthma, with (acute) exacerbation ×

J06.9 - Acute upper respiratory infection, unspecified ×

Prescriptions

Current Prescriptions

Albuterol HFA

2 puffs, Q4-6H PRN

Use as needed for shortness of breath.

Add New Prescription

Drug Name

Amoxicillin 500mg

Dosage

500mg

Frequency

TID

Instructions

Take three times daily with food for 7 days.

Add Prescription

Follow-up Plan

Patient to follow up with primary care physician in 3 days. Educated on warning signs for asthma exacerbation and when to seek urgent care. Provided written instructions for new medications.

CancelSave/Sign

Quick LinksResourcesCompany

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Search patients by Name, ID, o

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Critical Alerts

Patient Encounter

Clinical Notes

Brief summary of today's

Patient presented with

Follow-up scheduled i

!

Review and Confirm Changes

Please review the following validations before confirming the patient record update.

✓

All required fields completed, ensuring comprehensive data integrity.

⚠

Potential Penicillin Allergy Interaction detected. Confirm patient's allergy status before proceeding.

⚠

Serious Drug-drug interaction: Warfarin and Ibuprofen. Review medication regimen immediately.

📁

Archived patient notes from a previous visit are available for review.

Cancel

Confirm

Edit Encounter

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✓

Patient record updated successfully.

Return to Patient Record

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Search patients by Name, ID, o

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SYSTEM ALERTS

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Authorization

⚠️

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Network Connection Lost

It seems your device has lost connection to the server. Please check your internet connection. Data may be out of sync.

Save Draft Offline

Retry Connection

🌐

🐦

📺

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Detailed High-fidelity Wireframes

 **HealthFlowEHR**

Welcome Back

Access your secure patient records.

Username

Password

Login

[Forgot Password?](#)





[Dashboard](#)
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All Statuses

Search

Recent Patients



Eleanor Vance

DOB: 1985-04-12

ID: HFEHR-001-23

Active



Marcus Thorne

DOB: 1972-11-28

ID: HFEHR-002-23

Discharged



Sophia Chen

DOB: 1991-07-01

ID: HFEHR-003-23

Active



David Rodriguez

DOB: 1965-02-19

ID: HFEHR-004-23

Pending



Olivia White

DOB: 1998-09-05

ID: HFEHR-005-23

Active



William Green

DOB: 1954-03-15

ID: HFEHR-006-23

Discharged



Ava Martinez

DOB: 1989-12-21

ID: HFEHR-007-23

Active



James Taylor

DOB: 1978-06-08

ID: HFEHR-008-23

Pending

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DashboardPatientsScheduleSettings

Q Search patients by Name, ID, o

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John Doe

DOB: 1975-08-20

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Patient ID: HD75-08-20-001

Attending Physician: Dr. Emily Smith

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Critical Alerts

Anya Sharma

ID: HFEHR00123

DOB: 1985-04-15

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Cancel

Save/Sign

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Q Search patients by Name, ID, o

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Patient Encounter

Clinical Notes

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Cancel

Confirm

Edit Encounter

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Network Connection Lost

It seems your device has lost connection to the server. Please check your internet connection. Data may be out of sync.

Save Draft Offline

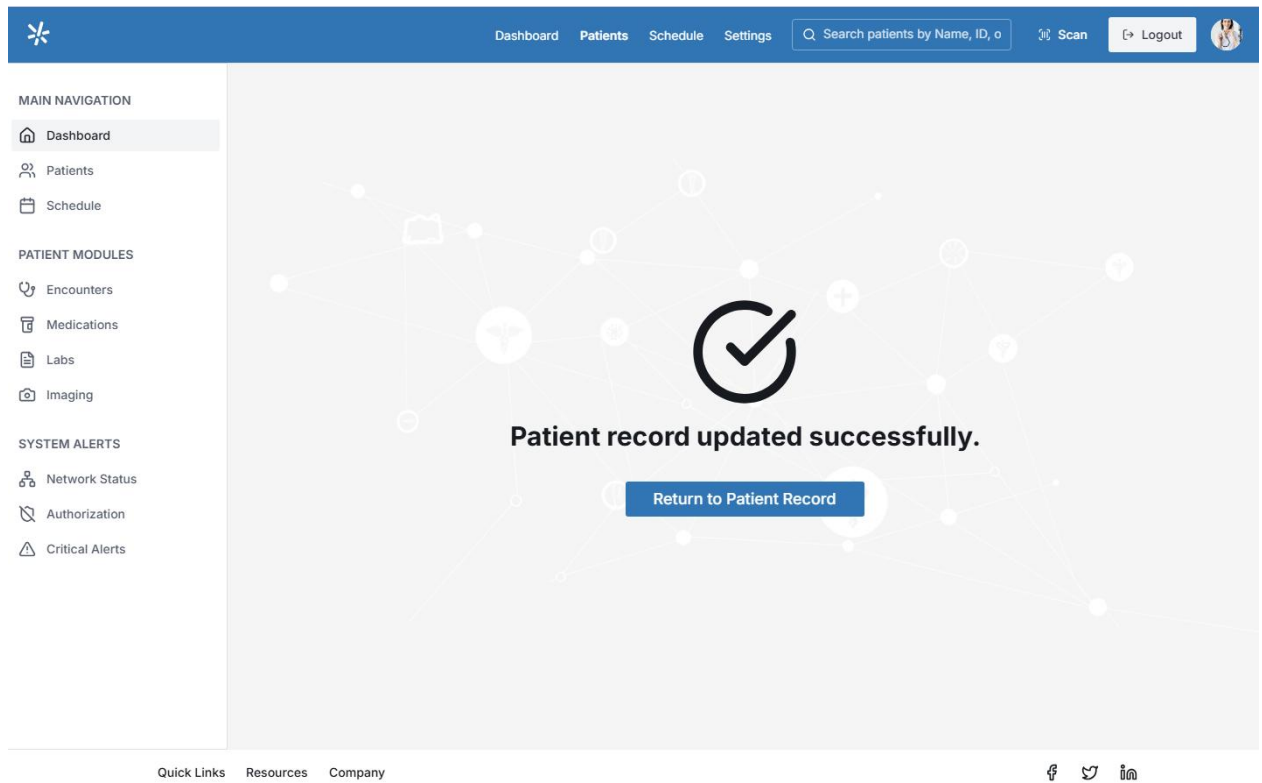
Retry Connection

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Detailed Use Case Scenario1: Register Patient

Use Case ID		RP001
Use Case Name		Register Patient
Summary		This use case describes how the receptionist registers a new patient into the Smart Healthcare System. The receptionist collects demographic details (name, age, NIC no,gender, address, contact), medical history, and assigns a unique patient ID. The system validates the data, stores it in the central database, and generates a digital health card (QR/Barcode) for the patient.
Priority		5 (1=lowest priority, 5=highest priority)
Preconditions		Receptionist is logged into the hospital system. Patient is new and not already registered. Registration module of the system is online.
Post Condition		Patient account is created in the system with a unique patient ID. Digital health card (QR/Barcode) is generated and assigned. Confirmation is displayed with audio and visual feedback.
Primary Actor		Receptionist
Trigger		A new patient arrives at the hospital and requests registration.
Main Success Scenario	Step	Action
	1	Receptionist opens the patient registration form.
	2	Receptionist enters personal details,NIC no, contact information, and initial medical history.
	3	System validates the entered data for completeness and correctness.
	4	System checks for duplicate records to avoid redundancy.
	5	If data is valid and no duplicate is found, the system stores patient data in the central database.
	6	System generates a unique patient ID and creates a digital health card (QR/Barcode).

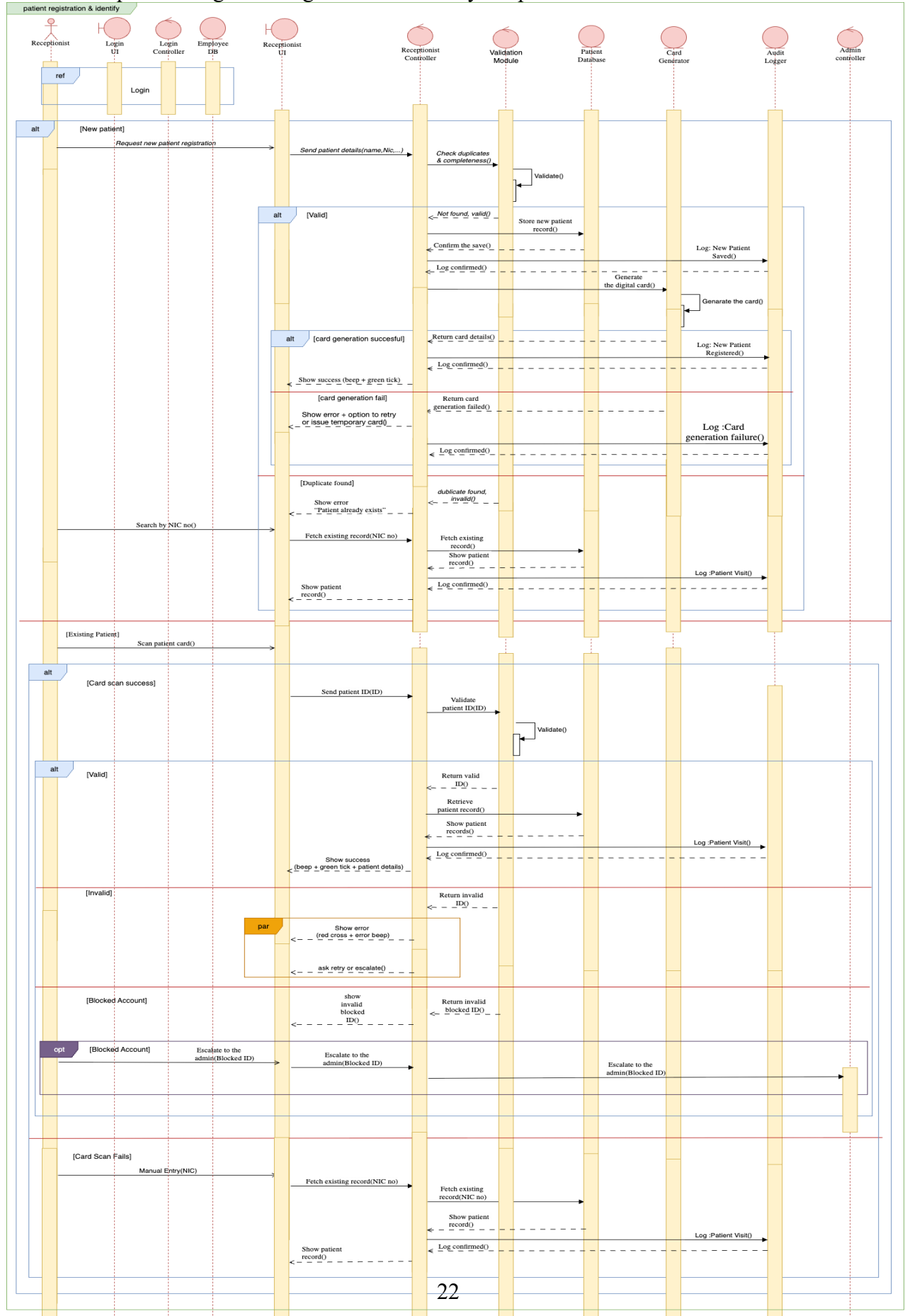
	7	System provides audio (success beep) and visual (confirmation message) feedback to indicate successful registration.
	8	System logs the registration event in the audit logger.
Extensions	Step	Branching Action
	3a	If required fields are missing → System prompts receptionist to re-enter details.
	4a	If duplicate patient record exists → Trigger Error Handling and notify receptionist to search patient record.
	4b	Receptionist search the patient record by NIC no and retrieve the patient record.
	7a	If card printing fails → Receptionist is prompted to retry or generate a temporary card.
Open Issues	1	Should patients be required to upload ID proof (NIC/Passport) during registration?
	2	Should biometric details (fingerprint/face ID) be included for future authentication?

Detailed Use Case Scenario 2: Verify Patient Identity

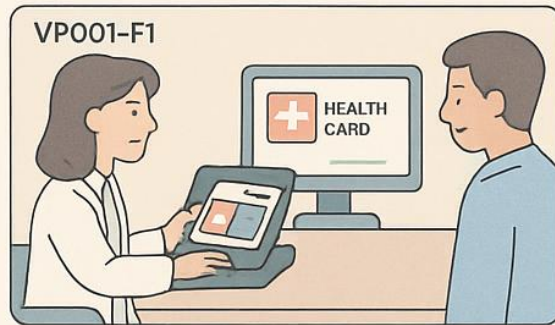
Use Case ID		VP001
Use Case Name		Verify Patient Identity
Summary		This use case explains how the receptionist verifies an existing patient's identity when they visit the hospital. The patient presents their digital health card (QR/Barcode), which is scanned by the system. The system validates the ID, retrieves the patient record, and confirms identity with audio and visual feedback. If scanning fails, manual entry or error handling is triggered.
Priority		5 (1=lowest priority, 5=highest priority)
Preconditions		Patient has a valid digital health card (QR/Barcode). Receptionist is logged into the system. Patient is already registered in the system database.
Post Condition		Patient record is successfully retrieved and displayed.

		Confirmation is given with audio and visual feedback.
Primary Actor		Receptionist
Trigger		Patient presents card at reception for identity verification.
Main Success Scenario	Steps	Action
	1	Receptionist scans the patient's card (QR/Barcode).
	2	System reads the card and validates the patient ID.
	3	System retrieves the patient's record from the database.
	4	System displays confirmation (patient name, ID, photo) along with audio success tone and visual success message.
	5	Receptionist confirms identity and proceeds with next hospital service (appointments/billing).
Extensions	Step	Branching Action
	1a	If card scan fails → Receptionist selects Manual Entry and enters Patient ID.
	2a	If patient ID is invalid → System provides error beep, red error message, and triggers Error Handling .
	3a	If duplicate or blocked account is found → Receptionist follows hospital policy for manual verification.
	4a	If system is offline → Receptionist uses offline patient lookup (backup).
Open Issues	1	Should the system store logs of all failed verification attempts for auditing?
	2	Should kiosks use the same verification workflow independently of receptionists?

Detailed Sequence Diagram : Register and indentify the patient



Use Case – Verify Patient Identity



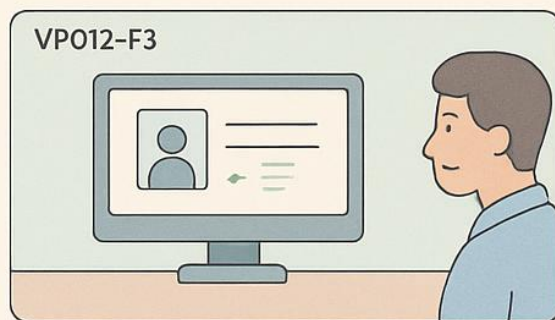
Scan Card

Receptionist scans the patient's digital health card.



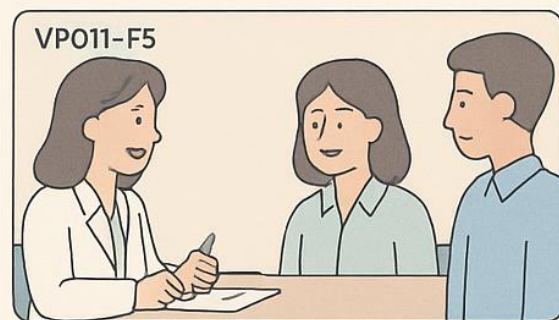
Validate Patient ID

System reads the card and validates the patient ID.



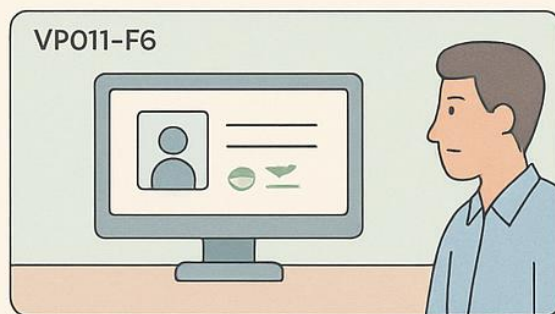
Retrieve Record

System retrieves patient details from the database.



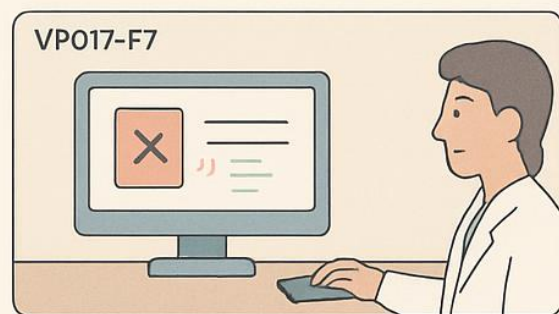
Confirm Identity

Receptionist confirms identity and continues with hospital service



Show Confirmation

System displays patient details with green tick and success beep.



Invalid ID (Error case)

If patient ID is invalid, system shows red cross and error beep.

Use Case RP001 – Register Patient



Open Registration Form

Receptionist opens patient registration form.

RP001-F1



Enter Patient Details

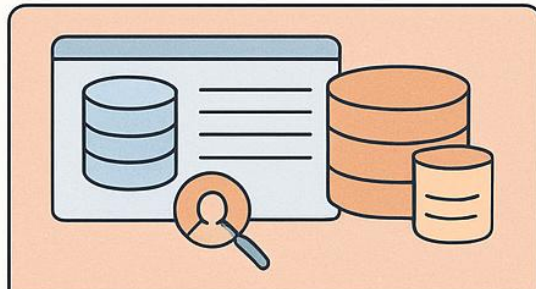
Receptionist enters patient demographics, NIC number, and medical history.



Validate Data

System checks completeness of details and prompts if required fields are missing.

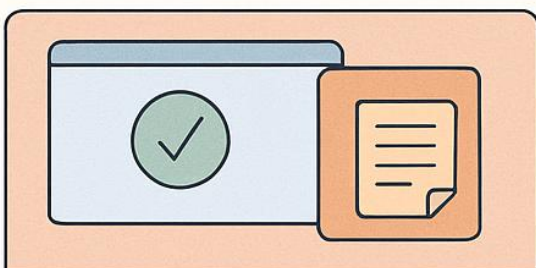
RP001-F3



Store Patient Record

System stores patient details in the database and assigns unique patient ID

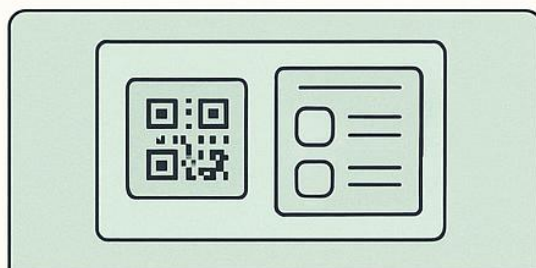
RP001-F5



Generate Digital Card

System shows confirmation with green tick and success beep

RP001-F7



Generate Digital Card

System shows confirmation with green tick and success beep

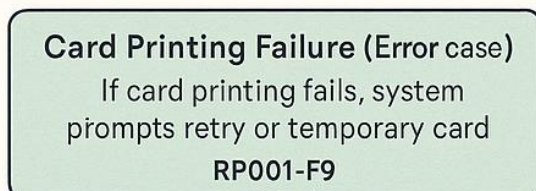
RP001-F8



Audit Logging

System logs the registration event for accountability

RP001-F8



Card Printing Failure (Error case)

If card printing fails, system prompts retry or temporary card

RP001-F9

Low-fidelity Wireframes

Welcome, Receptionist!

Username

Password

Login

 **HospitalSmartHealthcareSystem**

Logout

Verify Existing Patient

Scan Patient Card

Scan Patient Card

Use a QR/Barcode scanner to read the patient's card.

Enter NIC Number Manually

Verify NIC

Patient Record Details

Patient details will appear here after successful verification.

Register New Patient

Register New Patient

Patient Name

Enter full name

NIC Number

e.g., 1234567890123

Date of Birth

Gender

Select gender

Contact Information

Address

Enter street address, city, postal code

Contact Phone

e.g., 0771234567

Contact Email (Optional)

e.g., patient@example.com

Emergency & Guardian Details

Emergency Contact Name

Enter emergency contact's full name

Emergency Contact Phone

e.g., 0779876543

Guardian/Parent Name (if minor)

Enter guardian's full name

Medical History

Brief Medical History (Optional)

Describe any relevant medical conditions, past surgeries, etc.

☐ Known Allergies (e.g., medication, food, environmental)

☐ Currently on Medications

Cancel

Register Patient



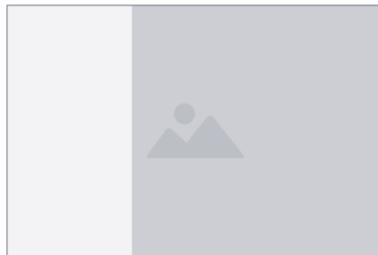
Registered successful.

Patient information verified.

View Patient Records

**Card Printing Failed**

The error has been logged for review.

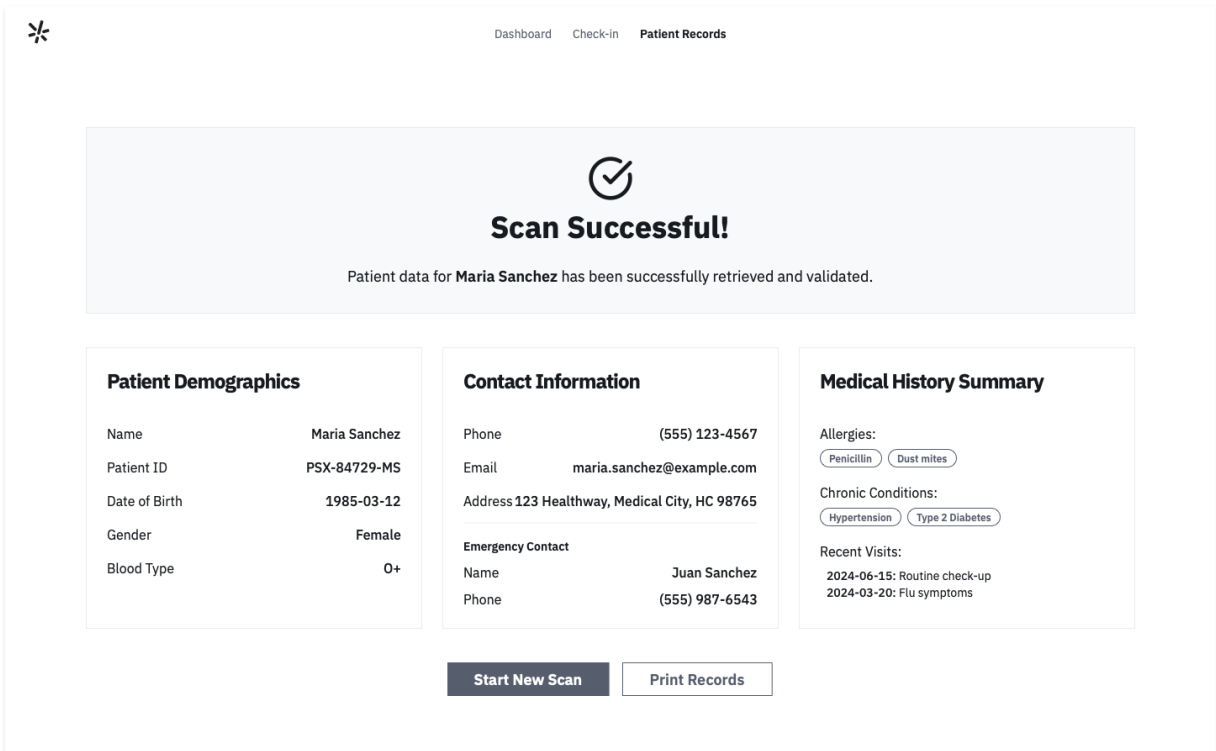
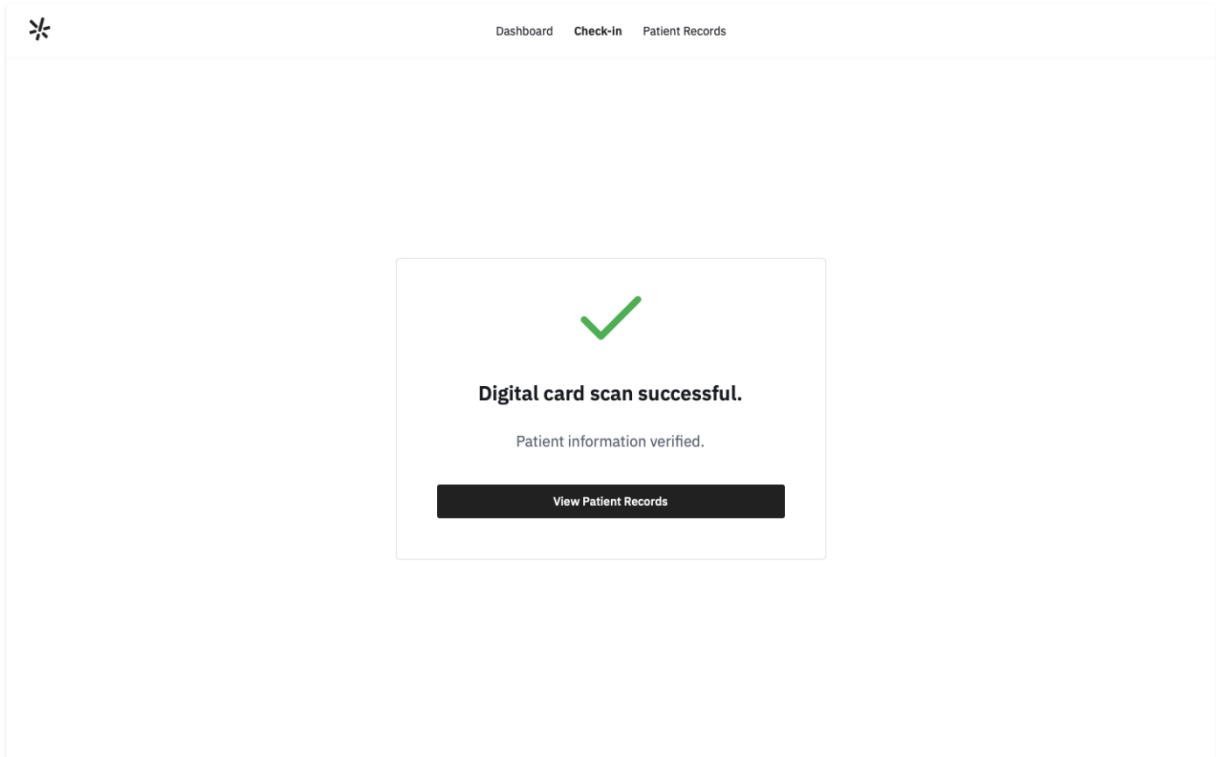
[Retry](#)[Issue Temporary Card](#)**Verify Existing Patient**

Scanning Digital ID...

Scan Information

Scan ID: SID-890123-A
Card Holder: John Doe
Status: [Scanning...](#)
Elapsed Time: 00:00

Actions[× Cancel Scan](#)[✓ Complete Manually](#)[📄 View Scan Logs](#)



Verify Existing Patient



x

Patient already exists

This patient's record is already in the system. You can search for the existing record or re-attempt card scanning.

[Search Existing Record](#)[Scan Card](#)**Scan Failed!**

The QR code could not be scanned or the entered ID is invalid. Please ensure the card is placed correctly or the ID is typed without errors.

[Retry Scan/Entry](#)[Manual NIC Entry](#)[Escalate to Admin](#)

Detailed High-fidelity Wireframes

Welcome, Receptionist!

Username

Password

Login



Logout

Verify Existing Patient

Scan Patient Card

Scan Patient Card

Use a QR/Barcode scanner to read the patient's card.

Enter NIC Number Manually

Verify NIC

Patient Record Details

Patient details will appear here after successful verification.

Register New Patient

Register New Patient

Patient Name

Enter full name

NIC Number

e.g., 1234567890123

Date of Birth

Gender

Select gender

Contact Information

Address

Enter street address, city, postal code

Contact Phone

e.g., 0771234567

Contact Email (Optional)

e.g., patient@example.com

Emergency & Guardian Details

Emergency Contact Name

Enter emergency contact's full name

Emergency Contact Phone

e.g., 0779876543

Guardian/Parent Name (if minor)

Enter guardian's full name

Medical History

Brief Medical History (Optional)

Describe any relevant medical conditions, past surgeries, etc.

☐ Known Allergies (e.g., medication, food, environmental)

☐ Currently on Medications

Cancel

Register Patient



Registered successful.

Patient information verified.

View Patient Records



Card Printing Failed

The error has been logged for review.

Retry

Issue Temporary Card



Scanning Digital ID...



Scan Information

Scan ID: SID-890123-A

Card Holder: John Doe

Status: Scanning...

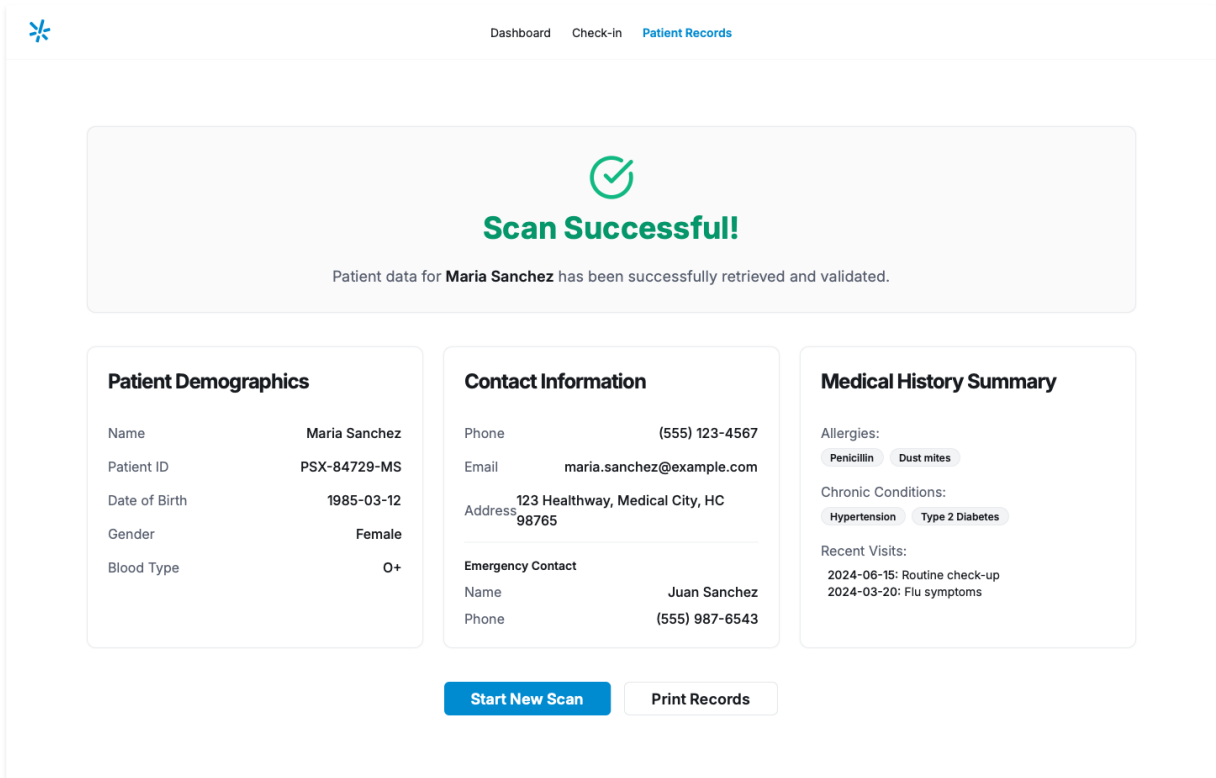
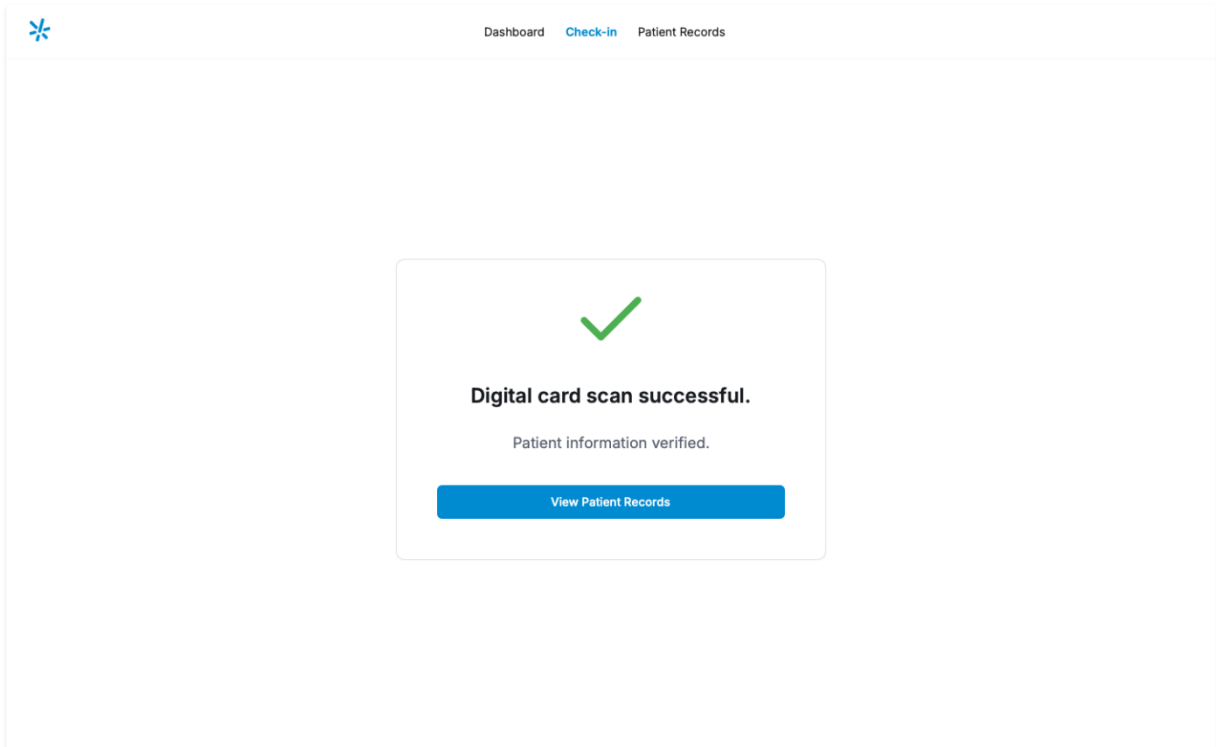
Elapsed Time: 00:00

Actions

Cancel Scan

Complete Manually

View Scan Logs



Verify Existing Patient



Patient already exists

This patient's record is already in the system.
You can search for the existing record or
re-attempt card scanning.

Search Existing Record

Scan Card



Scan Failed!

The QR code could not be scanned or the entered ID is invalid.
Please ensure the card is placed correctly or the ID is typed
without errors.

Retry Scan/Entry

Manual NIC Entry

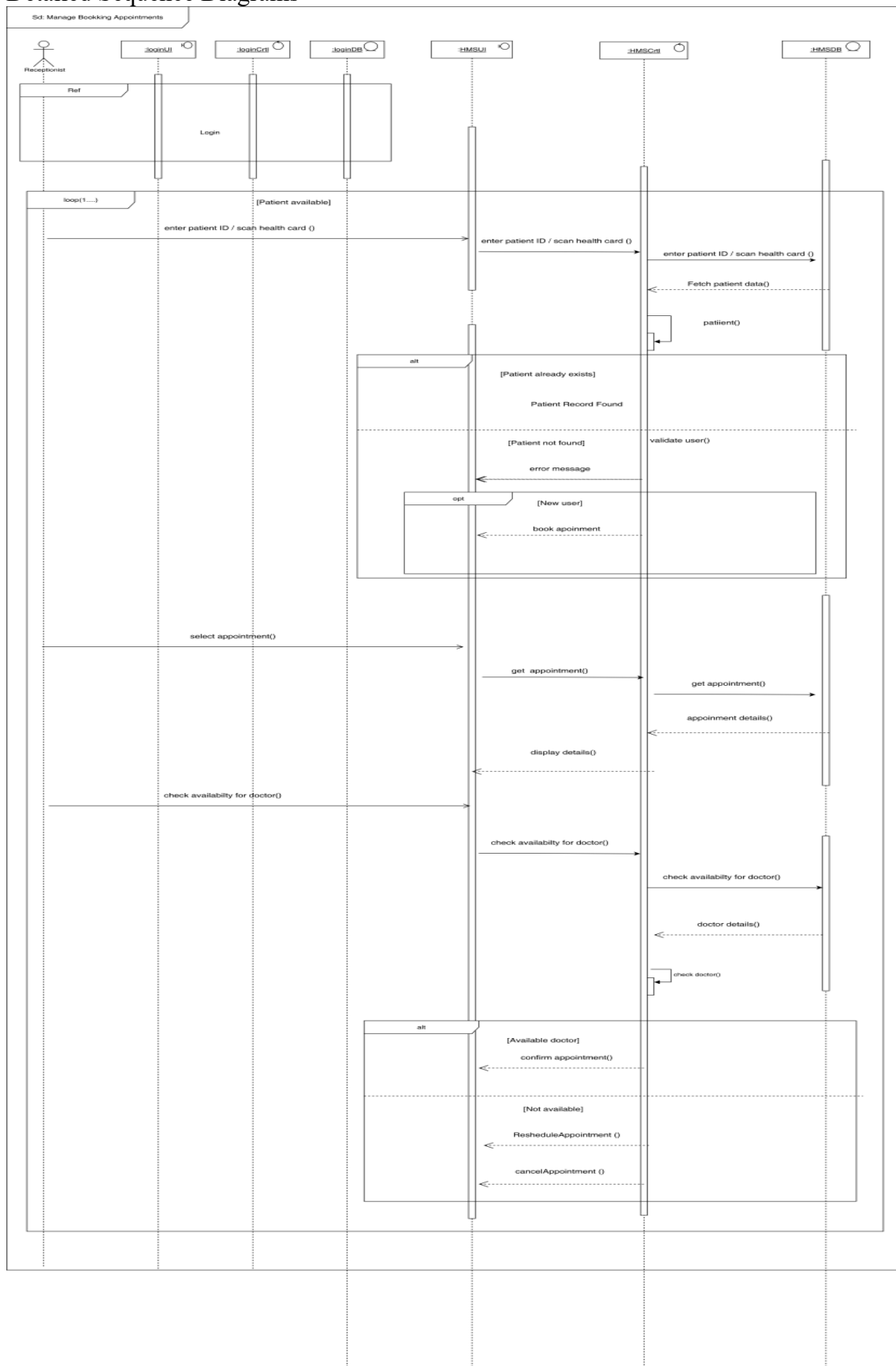
Escalate to Admin

Kajanan IT 23 3353 32
Detailed Use Case Scenarios

Use case ID	R_01	
Use case Name	Manage Appointment	
Main flow	To schedule, update, or cancel patient appointments efficiently.	
Priority	05	
Pre-condition	Patient has a valid digital health card/account in the system. Receptionist is logged into the hospital management portal.	
Post-condition	Appointment record is updated in the system. Patient and doctor schedules are synchronized.	
Primary Actor	Receptionist	
Main Success scenario	Step	Action
	01	Patient provides digital health card or patient ID at the reception.
	02	Receptionist logs into the system and enters/scans the patient ID.
	03	System retrieves and displays patient details with appointment history.
	04	Receptionist selects Manage Appointment option.
	05	Receptionist chooses an action: Book / Update / Cancel Appointment.
	06	If Book/Update , System checks doctor's availability.
	07	System displays available time slots.
	08	Receptionist and patient agree on a suitable time slot.
	09	Receptionist confirms the appointment in the system.
	10	System saves the appointment, updates records, and generates confirmation (print/email/SMS).
Extensions	02a	Invalid patient ID ,System shows an error ,Receptionist re-enters ID or asks patient to verify.
	02b	If login/ID is valid, display success message and show patient details.

	03a	If patient record not found, show popup to register new patient.
	03b	If patient already exists in the system, notify receptionist.
	05a	If patient requests to cancel appointment, update system and free slot.
	06a	If no doctor slots available, suggest next available time or alternative doctor.
	10a	If notification service fails, log appointment and retry later.

Detailed Sequence Diagrams



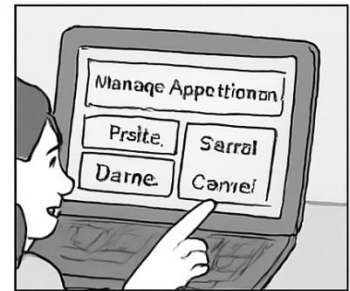
Storyboards



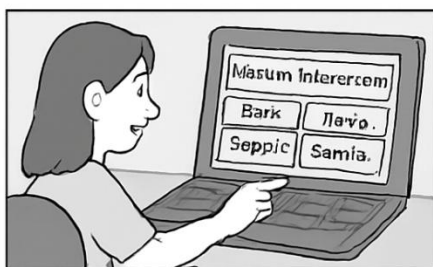
Logs 'logs into the hospital system



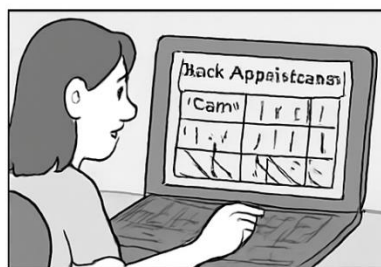
searches for patient by ID card



clicks 'Manage Appointment'



User: clicks 'Manage Appointment'
System: options displayed



User: selects a doctor's available time slot
System: available



User: provides confirmation
System: confirmation &



1. Reddvost



1. Manage Details



Booking time slots



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Printination



USE: redicuts

Low-fidelity Wireframes



PATIENT INFORMATION

Patient Name

Jane Smith

Patient ID

PID4567

Age

42

Contact Number

987-654-3210

Medical History

Allergies to Penicillin. Recent flu.
Previous heart condition.

Appointment Management

Book New Appointment

Reschedule Appointment

Cancel Appointment

Current Appointment Details

Appointment Type

General Check-up

Doctor

Dr. Sarah Lee

Date

April 25, 2024

Time

10:00 AM - 10:30 AM

Patient

Jane Smith

Recent Appointments

Patient	Doctor	Date	Time	Status
Jane Smith	Dr. Sarah Lee	Apr 25, 2024	10:00 AM	Confirmed
Robert Johnson	Dr. Emily White	Apr 24, 2024	02:00 PM	Completed
Alice Brown	Dr. John Doe	Apr 23, 2024	11:00 AM	Canceled
Michael Davis	Dr. Sarah Lee	Apr 22, 2024	09:00 AM	Confirmed
Emily Wilson	Dr. Emily White	Apr 21, 2024	03:30 PM	Pending

Doctor Schedule

Dr. Sarah Lee

April 2024

Mon	Tue	Wed	Thu	Fri	Sat	Sun
1 9:00 AM 10:00 AM 11:00 AM	2 Busy	3 1:00 PM 2:00 PM	4 9:30 AM 10:30 AM	5 Busy	6 Weekend	7 Weekend
8 1:00 PM 2:00 PM	9 9:00 AM 10:00 AM	10 Busy	11 11:00 AM 1:00 PM	12 9:00 AM 10:00 AM	13 Weekend	14 Weekend
15 Busy	16 2:00 PM 3:00 PM	17 9:00 AM 10:00 AM	18 1:00 PM 2:00 PM	19 Busy	20 Weekend	21 Weekend
22 9:00 AM 10:00 AM	23 Busy	24 1:00 PM 2:00 PM	25 9:30 AM 10:30 AM	26 Busy	27 Weekend	28 Weekend
29 1:00 PM 2:00 PM	30 9:00 AM 10:00 AM					

Print Confirmation

Email Confirmation

SMS Reminder

39

Detailed High-fidelity Wireframes



PATIENT INFORMATION

Patient Name

Jane Smith

Patient ID

PID4567

Age

42

Contact Number

987-654-3210

Medical History

Allergies to Penicillin. Recent flu.
Previous heart condition.

Appointment Management

Book New Appointment

Reschedule Appointment

Cancel Appointment

Current Appointment Details

Appointment Type

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Emily Wilson	Dr. Emily White	Apr 21, 2024	03:30 PM	Pending

Logout



Doctor Schedule

Dr. Sarah Lee

April 2024

Mon	Tue	Wed	Thu	Fri	Sat	Sun
1 9:00 AM 10:00 AM	2 Busy	3 1:00 PM 2:00 PM	4 9:30 AM 10:30 AM	5 Busy	6 Weekend	7 Weekend
8 1:00 PM 2:00 PM	9 9:00 AM 10:00 AM	10 Busy	11 11:00 AM 1:00 PM	12 9:00 AM 10:00 AM	13 Weekend	14 Weekend
15 Busy	16 2:00 PM 3:00 PM	17 9:00 AM 10:00 AM	18 1:00 PM 2:00 PM	19 Busy	20 Weekend	21 Weekend
22 9:00 AM 10:00 AM	23 Busy	24 1:00 PM 2:00 PM	25 9:30 AM 10:30 AM	26 Busy	27 Weekend	28 Weekend
29 1:00 PM 2:00 PM	30 9:00 AM 10:00 AM					

Print Confirmation

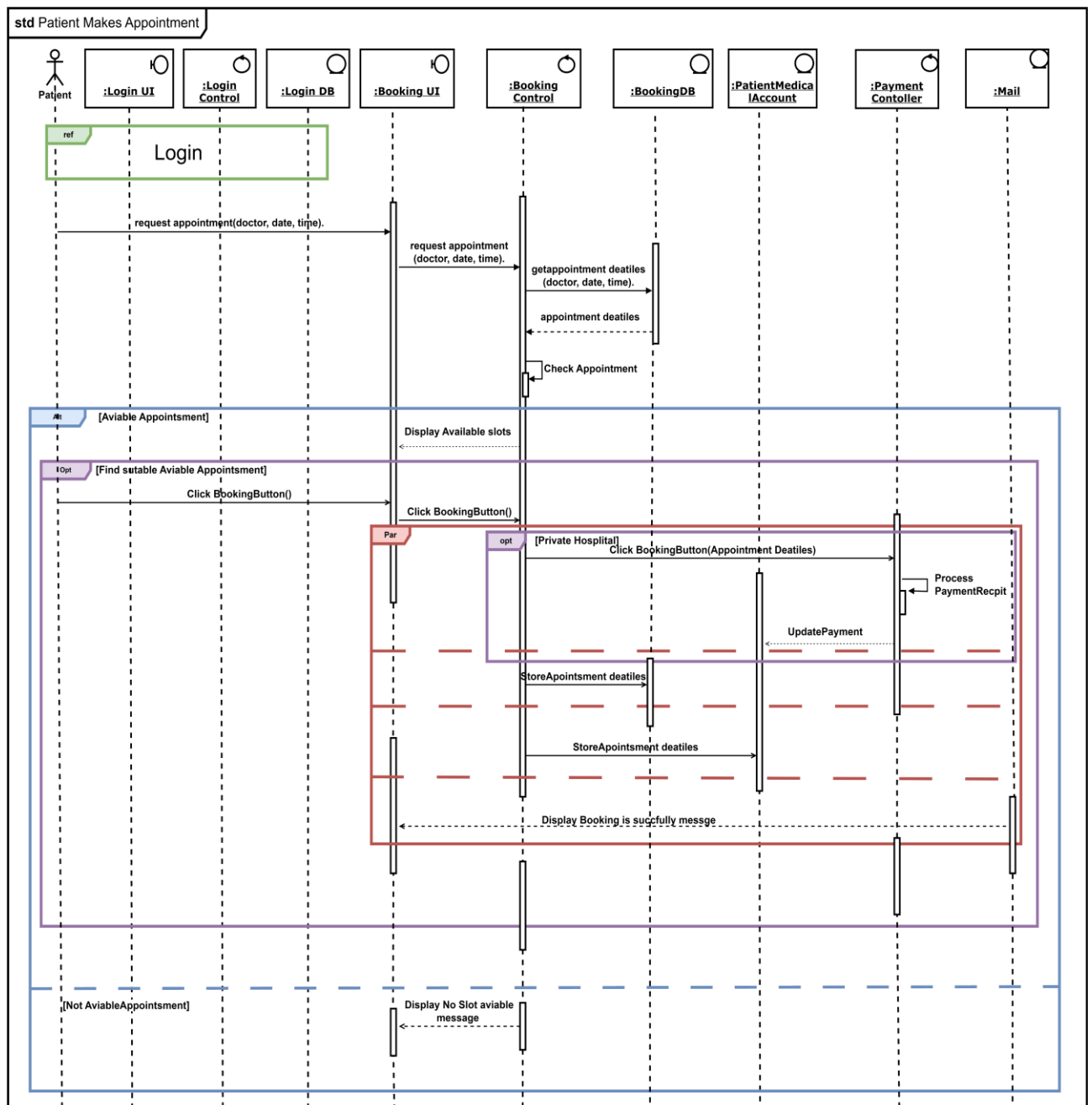
Email Confirmation

SMS Reminder

Use Case ID:	VP004
Use Case Name:	Patient Appointment Booking
Summary:	This use case describes how a patient can book, view, reschedule, or cancel an appointment using the digital health system via web portal, mobile app, or hospital kiosk. The system confirms availability, updates patient records, and sends notifications.
Priority:	5 (1=lowest, 5=highest)
Preconditions:	1. Patient has a valid digital health card and registered account. 2. System is online and accessible. 3. Patient's demographic and medical history are stored in the system.
Post Conditions:	1. Appointment is successfully booked, rescheduled, or cancelled. 2. Patient's records are updated in real-time. 3. Notifications are sent to patient and hospital staff.
Primary Actor:	Patient
Secondary Actors:	Hospital Staff, Digital Health System
Trigger:	Patient logs into the system and chooses to book, reschedule, or cancel an appointment.
Main Success Scenario (Steps/Action):	1. Patient logs into the system via web, mobile app. 2. Patient selects "Book Appointment". 3. System displays available doctors, dates, and time slots. 4. Patient chooses doctor, date, and time. 5. System verifies slot availability. 6. System confirms booking. 7. System updates patient's record. 8. System sends confirmation via SMS, email, or app notification.
Extensions / Alternate Flows:	Reschedule Appointment: 1. Patient selects existing appointment. 2. Patient chooses new date and time. 3. System verifies and confirms rescheduling.

	4. Records updated and notifications sent. Cancel Appointment: 1. Patient selects appointment to cancel. 2. System requests confirmation. 3. Appointment is cancelled and removed. 4. 4. Records updated and notifications sent.	
Exception Flows:	STEP	
	5 a	If Selected slot unavailable -> System shows alternative available slots , Patient selects a new slot or exits without booking.
	6a	If system downtime occurs → System notifies the patient. , System suggests retrying later or contacting hospital staff directly.
Open Issues:	1. Should the system allow push notifications for reminders 2. Should patients be able to cancel/reschedule after a cut-off time?	

Detailed Sequence Diagrams



Storyboards



Patient logging into hospital system via laptop or



Select Book Appointment



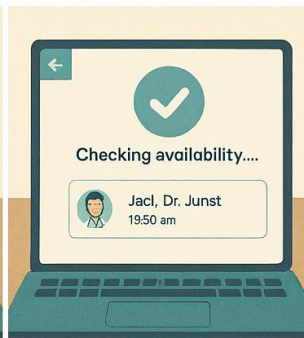
System shows a list of available doctors profiles



Conelictet doctor, dateo, time slot withment thm



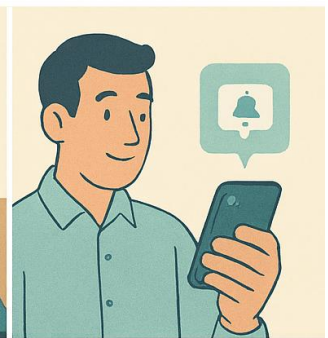
System confirms book-ring amnioctio



Appointment confirmed with success message



Patient record screen updates automatically showing

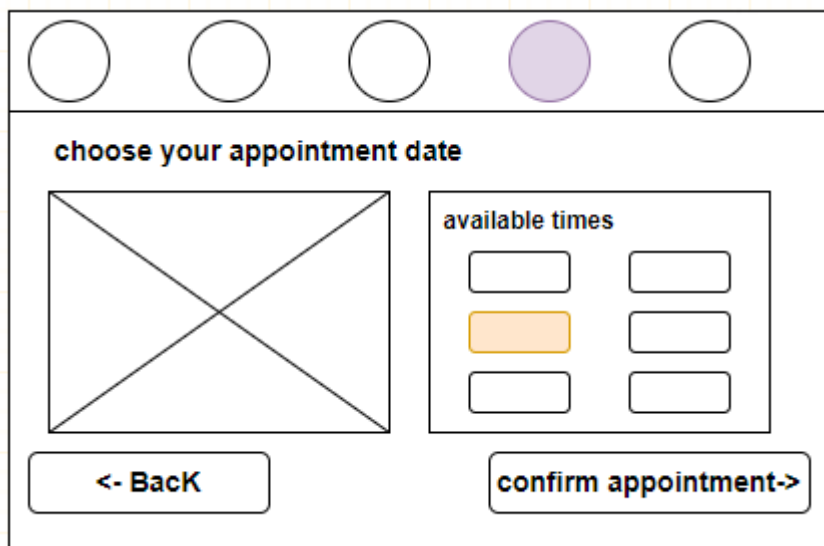


Patient receives notification via smartphone

Low-fidelity Wireframes

A low-fidelity wireframe for a screen titled "choose the department". At the top, there is a horizontal bar containing five circles; the second circle from the left is filled with a light purple color, indicating the current step. Below this bar, the text "choose the department" is displayed. Underneath the text, there are three rectangular boxes, each containing a large 'X' and a wavy line below it, representing placeholder images or text. At the bottom of the screen, there are two buttons: "<- Back" on the left and "Next->" on the right.

A low-fidelity wireframe for a screen titled "choose the doctor". At the top, there is a horizontal bar containing five circles; the third circle from the left is filled with a light purple color, indicating the current step. Below this bar, the text "choose the doctor" is displayed. Underneath the text, there are two rectangular input fields, each containing a wavy line, representing text input areas. At the bottom of the screen, there are two buttons: "<- Back" on the left and "Next->" on the right.



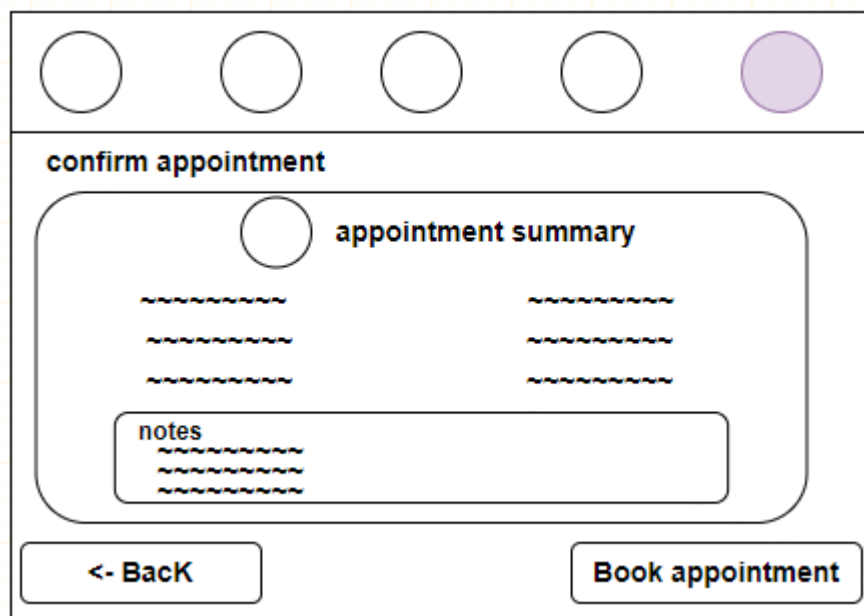
A window titled "choose your appointment date" with a title bar containing five buttons, the fourth of which is highlighted in purple. The main content area is divided into two sections. On the left is a large square box with a diagonal 'X' across it. On the right, under the heading "available times", is a 2x3 grid of input fields; the middle-left field is highlighted in orange. At the bottom are two buttons: "<- Back" on the left and "confirm appointment->" on the right.

choose your appointment date

available times

<- Back

confirm appointment->



A window titled "confirm appointment" with a title bar containing five buttons, the fifth of which is highlighted in purple. The main content area features a rounded rectangle containing an "appointment summary" section with a circular icon and wavy lines representing text. Below this is a "notes" section with a text input field and wavy lines. At the bottom are two buttons: "<- Back" on the left and "Book appointment" on the right.

confirm appointment

appointment summary

notes

<- Back

Book appointment

Detailed High-fidelity Wireframes

1 Select Hospital

2 Choose Department

3 Choose Doctor

4 Date & Time

5 Confirm

Choose Department



Cardiology

Heart and cardiovascular care

5 doctors available



Neurology

Brain and nervous system

3 doctors available



Orthopedics

Bones, joints, and muscles

4 doctors available

← Back

Continue to Doctors →

1 Select Hospital

2 Choose Department

3 Choose Doctor

4 Date & Time

5 Confirm

Choose Doctor

DS

Dr. Sarah Johnson

Cardiologist • 15 years experience

★★★★★ 4.9/5 (127 reviews)

Available Today

Next: 2:30 PM

MB

Dr. Michael Brown

Cardiologist • 12 years experience

★★★★★ 4.8/5 (89 reviews)

Available Tomorrow

Next: 9:00 AM

← Back

Select Date & Time →

1 Select Hospital

2 Choose Department

3 Choose Doctor

4 Date & Time

5 Confirm

Pick Date & Time

Select Date

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	31	1	2	3
4	5	6	7	8	9	10
11	12	13	14	15		

Available Times

9:00 AM

9:30 AM

10:00 AM

10:30 AM

2:00 PM

2:30 PM

3:00 PM

3:30 PM

← Back

Confirm Appointment →

1 Select
Hospital

2 Choose
Department

3 Choose
Doctor

4 Date &
Time

5 Confirm

Confirm Appointment



Appointment Summary

Hospital

City Hospital - Downtown Medical Center

Department

Cardiology

Doctor

Dr. Sarah Johnson

Date

March 11, 2024

Time

10:00 AM

Duration

30 minutes

Important Notes

- Please arrive 15 minutes early for check-in
- Bring your insurance card and ID
- Fasting required 12 hours before appointment

← Back

Book Appointment ✓

Appendix

1. Register Patient storyboard (RP001) Prompt

Create a **calm card-style storyboard** for **Use Case RP001 – Register Patient**, using soft pastel color blocks (blue, green for success steps; orange/red for error steps). Each card must have:

- A bold number (RP001-F1, RP001-F2 ...).
- A short clear title.
- A simple flat illustration.
- A one-line caption (clear, minimal).

Storyboard steps:

1. **RP001-F1: Open Form** – Receptionist opens patient registration form.
2. **RP001-F2: Enter Details** – Receptionist enters patient information (name, NIC, contact, medical history).
3. **RP001-F3: Validate Data** – System validates data, prompts if required fields are missing.
4. **RP001-F4: Check Duplicate** – System checks duplicates; if found, receptionist retrieves existing record.
5. **RP001-F5: Store Record** – System stores patient data and generates unique patient ID.
6. **RP001-F6: Generate Card** – System generates digital QR/Barcode health card.
7. **RP001-F7: Confirm Success** – System shows confirmation with green tick and beep.
8. **RP001-F8: Audit Logging** – System logs the registration event.
9. **RP001-F9: Card Printing Failure (Error)** – If printing fails, receptionist retries or issues a temporary card.

2. Register Patient storyboard (RP001) Prompt

Create a calm, card-style storyboard for **Use Case VP001 – Verify Patient Identity**. Use soft pastel color blocks (blue/green for normal steps, orange/red for errors). Each card should include:

- An index (VP001-F1, VP001-F2 ... VP001-F9).
- A short clear title.
- A flat illustration.
- A one-line caption with the following text:
- **VP001-F1: Scan Card** – Caption: “Receptionist scans the patient’s digital health card using the hospital system.”
- **VP001-F2: Validate Patient ID** – Caption: “System reads the card and validates the patient ID against the database.”

- **VP001-F3: Retrieve Record** – Caption: “System retrieves patient details from the database if the ID is valid.”
- **VP001-F4: Show Confirmation** – Caption: “System displays patient details (name, ID, photo) with a green tick and success beep.”
- **VP001-F5: Confirm Identity** – Caption: “Receptionist confirms the patient’s identity and continues with hospital services (appointments, billing, treatment).”
- **VP001-F6: Card Scan Failure (Error case)** – Caption: “If the card scan fails, the receptionist switches to manual entry by typing the NIC number.”
- **VP001-F7: Invalid ID (Error case)** – Caption: “If the patient ID is invalid, the system displays a red cross, plays an error beep, and prompts retry or escalation.”
- **VP001-F8: Blocked/Duplicate Account (Error case)** – Caption: “If the account is blocked or a duplicate is detected, the receptionist follows hospital policy for manual verification or escalation.”
- **VP001-F9: Offline Lookup (Error case)** – Caption: “If the system is offline, the receptionist retrieves patient records from the offline backup system until normal service is restored.”

3. Patient Appointment Booking Prompt

Frame 1 – Patient Login

- **Scene:** Patient sits at a laptop or holds a smartphone.
- **UI Elements:** Login screen with fields for digital health card number / credentials.
- **Visual Cues:** Digital health card icon, login button highlighted.
- **Text/Label:** “Patient logs into the hospital system using a digital health card.”

Frame 2 – Select 'Book Appointment'

- **Scene:** Patient dashboard displayed after login.
- **UI Elements:** Menu with options like ‘My Records’, ‘Book Appointment’, ‘Notifications’.
- **Visual Cues:** ‘Book Appointment’ button highlighted, cursor or finger tapping it.
- **Text/Label:** “Patient selects ‘Book Appointment’ from the dashboard menu.”

Frame 3 – Available Doctors List

- **Scene:** System shows a list of doctors.
- **UI Elements:** Doctor profiles with photo, specialty, available dates, clickable time slots.
- **Visual Cues:** Grid/list layout, clickable time slots with hover effect.
- **Text/Label:** “System displays available doctors, their specialties, and open appointment times.”

Frame 4 – Visual Calendar Highlighted

- **Scene:** Calendar view pops up for scheduling.
- **UI Elements:** Calendar with available dates highlighted.
- **Visual Cues:** Selected date highlighted in bright color.
- **Text/Label:** “Patient views calendar and selects preferred date for appointment.”

Frame 5 – Select Doctor, Date, and Time

- **Scene:** Doctor profile with selected time slot highlighted.
- **UI Elements:** Doctor’s card, chosen date/time visually emphasized.
- **Visual Cues:** Bright highlight on chosen slot, checkmark icon.
- **Text/Label:** “Patient chooses doctor, date, and time. Selected options are highlighted.”

Frame 6 – System Verifies Slot

- **Scene:** Loading or verification animation.
- **UI Elements:** Spinner animation or progress bar.
- **Visual Cues:** ‘Checking availability...’ message, subtle animation.
- **Text/Label:** “System verifies appointment slot availability.”

Frame 7 – Booking Confirmation

- **Scene:** Success message displayed.
- **UI Elements:** Confirmation card with doctor, date, time, and celebratory icons.
- **Visual Cues:** Green checkmark, confetti, smiling doctor icon.
- **Text/Label:** “Booking confirmed! Appointment details shown with celebratory icons.”

Frame 8 – Patient Record Update & Notification

- **Scene:** Patient record screen and notification pop-up.
- **UI Elements:** Appointment appears in patient records; smartphone/email notification pops up.
- **Visual Cues:** Notification icon, alert message, highlighted new appointment in records.
- **Text/Label:** “Patient record updates automatically. Notification received via smartphone or email.”

4. Doctor Accesses Patient Card & Updates Record prompt

Create a calm, card-style storyboard for **Use Case EP-DOC-01 – Doctor Accesses Patient Card & Updates Record**, using soft pastel color blocks (blue/green for normal steps, orange/red for error cases). Each card should include:

- An index (EP-DOC-01-F1, F2 ...).
- A short clear title.
- A flat illustration.

- A one-line caption.

Storyboard frames:

1. **EP-DOC-01-F1: Login** – Caption: “Doctor logs into the hospital EHR system with valid credentials.”
2. **EP-DOC-01-F2: Find Patient** – Caption: “Doctor scans patient’s health card or searches by NIC/ID.”
3. **EP-DOC-01-F3: Confirm Patient** – Caption: “System displays patient name, DOB, and photo to prevent misidentification.”
4. **EP-DOC-01-F4: Open Record** – Caption: “Doctor opens the patient’s medical record.”
5. **EP-DOC-01-F5: View Full Record** – Caption: “System shows demographics, history, allergies, medications, labs, imaging.”
6. **EP-DOC-01-F6: Open Encounter** – Caption: “Doctor navigates to current encounter or creates a new one if needed.”
7. **EP-DOC-01-F7: Enter Updates** – Caption: “Doctor adds notes, vitals, diagnoses, orders, prescriptions, and follow-up plan.”
8. **EP-DOC-01-F8: Save/Sign Updates** – Caption: “Doctor reviews updates and clicks Save/Sign.”
9. **EP-DOC-01-F9: Validation Checks** – Caption: “System checks required fields, drug interactions, and requests confirmation if needed.”
10. **EP-DOC-01-F10: Save & Audit** – Caption: “System saves changes, writes audit log, and updates downstream services (pharmacy, lab).”
11. **EP-DOC-01-F11: Confirmation** – Caption: “System shows success message and updated record view.”

1.1.1 Error/Extension Frames

12. **EP-DOC-01-F12: Scan Error** – Caption: “If card not readable, system prompts retry or manual search.”
13. **EP-DOC-01-F13: Record Not Found** – Caption: “If no patient found, system shows ‘Record not found’ and offers re-search.”
14. **EP-DOC-01-F14: Misidentification Warning** – Caption: “If mismatch detected, system denies access pending verification.”
15. **EP-DOC-01-F15: Authorization Error** – Caption: “If doctor lacks permissions, system shows access denied message.”
16. **EP-DOC-01-F16: Missing Required Data** – Caption: “System prevents saving until mandatory fields like diagnosis are completed.”
17. **EP-DOC-01-F17: Drug Interaction Alert** – Caption: “If allergy or drug interaction detected, system shows safety alert.”
18. **EP-DOC-01-F18: Offline Draft Save** – Caption: “If network fault occurs, system stores a temporary draft and syncs later.”
19. **EP-DOC-01-F19: Aborted Save** – Caption: “If doctor cancels, system discards changes or stores as draft.”
20. **EP-DOC-01-F20: Validation Error** – Caption: “If values are invalid, system highlights fields and prevents save.”
21. **EP-DOC-01-F21: Save Conflict** – Caption: “If another user updated the record, system shows merge/conflict resolution dialog.”

22. **EP-DOC-01-F22: Downstream Service Failure** – Caption: “If pharmacy interface is down, system saves locally and queues retry.”

Thank you.